

AICTE EDUSKILL VIRTUAL INTERNSHIP

In-Memory Cache with TTL Lite

Student Credentials

Manish Kumar

Enrollment No:2410030747

Program & Semester: B.Tech CSE (5th)

AICTE & EDUSKILLS VIRTUAL INTERNSHIP

IN-MEMORY CACHE WITH

An Engineering Capstone Presentation on High-Performance In-Memory Data Storage, Automated TTL Eviction, and Real-Time Telemetry.

TTLITE

PRESENTED BY
Manish Kumar
(2410030747)

DEGREE & INSTITUTION
B.Tech CSE, IILM
University

INDUSTRY MENTOR
Ms. Sagarabala
Behera

Internship Background

EduSkills Academy

Pioneering EdTech Ecosystem: EduSkills Academy collaborates closely with national bodies like AICTE and top tier industry partners to deliver production-grade technological training and internship opportunities.

Core Focus: Practical exposure to emerging technologies, closing academic-industry gaps, and instilling standard engineering best practices.

Program Credentials

Domain: MERN Full Stack Development + Engineering Capstone.

Supervision & Mentorship: Guided by industry mentor Ms. Sagarabala Behera and faculty members from the School of Computer Science & Engineering, IILM University.

Format: Structured 8-week intensive virtual internship curriculum with milestone assessments.

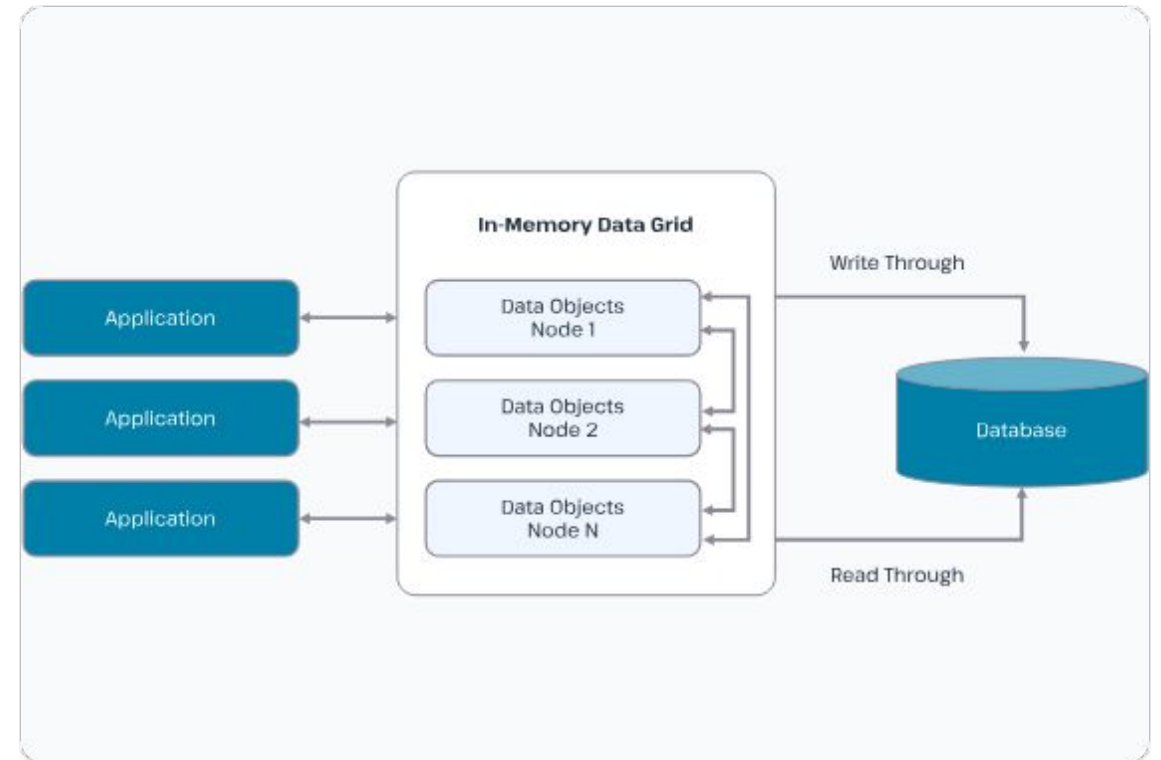
Problem Statement & Need

 **Database Bottlenecks:** Repeated queries for unchanged records strain database I/O, leading to severe latency spikes under high concurrency.

 **Latency Penalties:** Disk-based lookups and network roundtrips drastically degrade response times for real-time applications.

 **Stale Data Risks:** Without automated lifecycle management (TTL), cache entries persist indefinitely, causing memory bloat and consistency issues.

 **The Solution:** A lightweight, deterministic In-Memory Cache supporting SET/GET, auto-invalidation, and real-time hit/miss metrics.



Key Project Objectives



In-Memory Ops

Engineer high-efficiency SET, GET, and DELETE operations using optimized in-memory hash structures with near $O(1)$ time complexity.



TTL Invalidation

Implement proactive and reactive Time-To-Live (TTL) expiration algorithms to automatically discard stale memory records without leaks.



Telemetry & Metrics

Track real-time performance telemetry including cache hits, cache misses, hit ratios, and memory capacity utilization reports.

Technologies & Tooling

Domain / Layer	Technology / Tool	Role in Project & Internship
Core Language	JavaScript (ES6+) / Node.js	In-memory data structures, async timers, runtime environment
Backend & APIs	Express.js, RESTful Architecture	Routing, request handling, endpoint orchestration, JWT auth
Data & Persistence	MongoDB & Mongoose ODM	Primary persistent storage layer and schema modeling
Frontend Stack	React.js, Redux Toolkit, Tailwind CSS	Reactive user interfaces, centralized state, responsive UI
DevOps & Quality	Git, GitHub, VS Code	Version control, automated linting, test suites and validation

System Architecture

The engine orchestrates data validity, rapid retrieval, and telemetry through specialized decoupled modules:

SET Module

Stores key-value pairs with customizable TTL timestamps.

GET Module

Checks expiry, updates hit/miss counters, returns data.

TTL Engine

Automated eviction timer sweeps expired records.

DELETE Module

Allows instant programmatic or manual key purging.

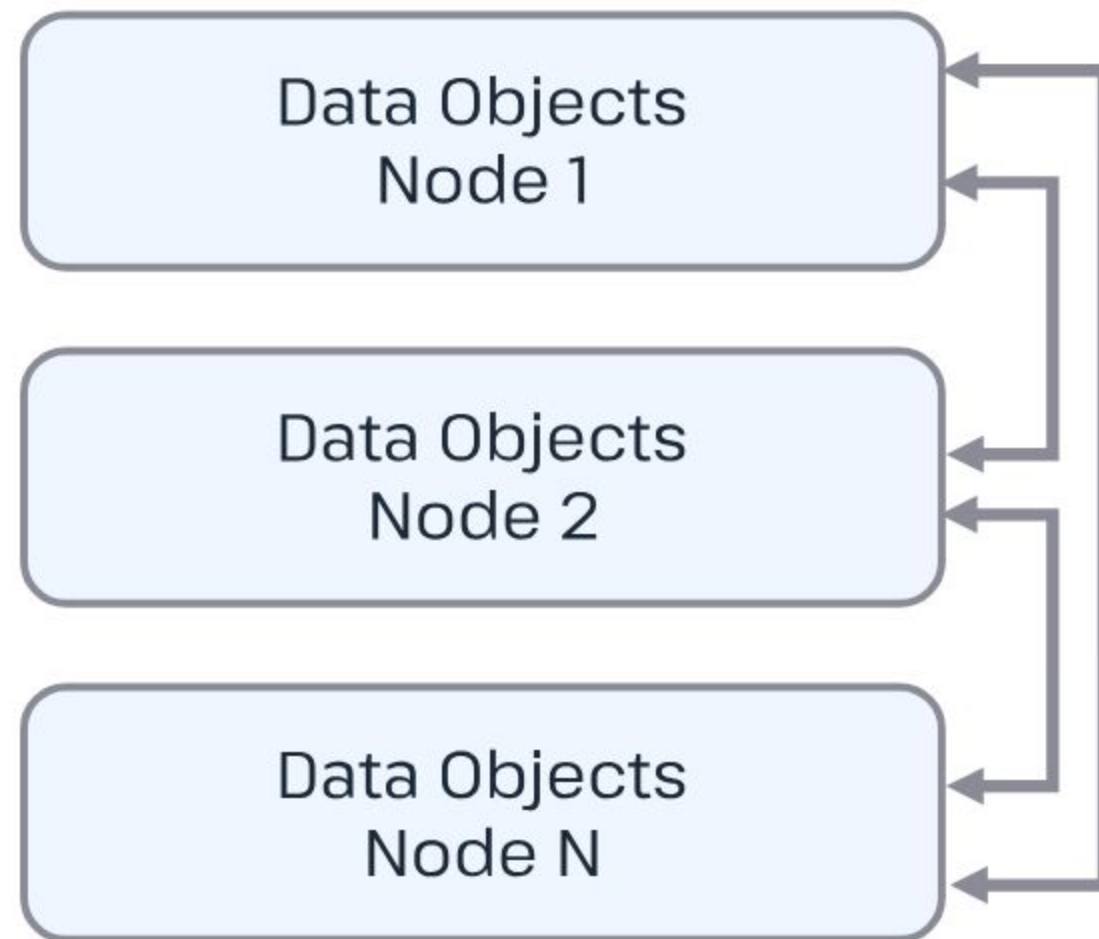
Stats Engine

Maintains atomic counters for hit ratio analytics.

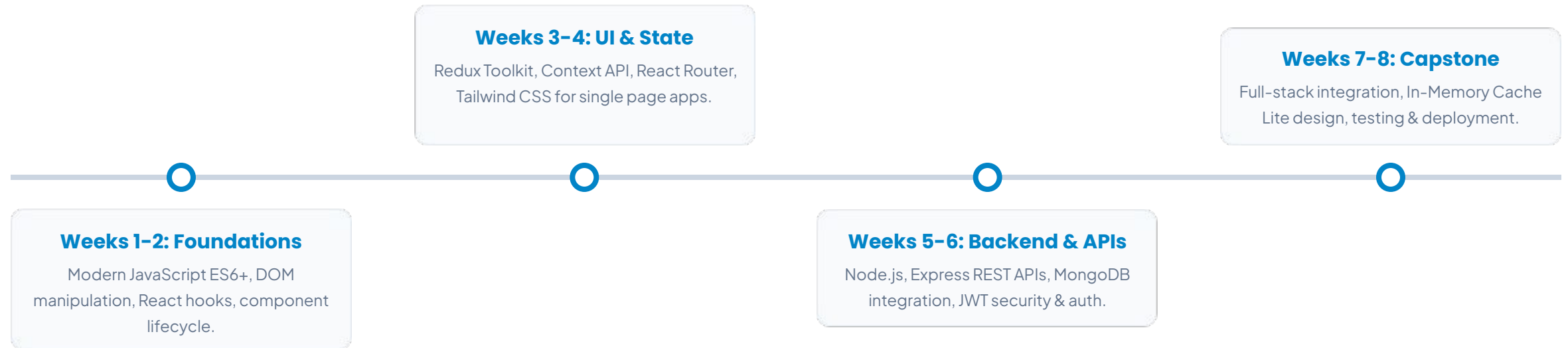
Report Generator

Outputs structured summary of cache performance.

In-Memory Data Grid



8-Week Internship Roadmap



Development Methodology



Phase 1 & 2: Core + TTL

Implemented primary key-value store in memory, followed by timestamp-based TTL eviction logic and scheduled cleanup sweeps.



Phase 3 & 4: Ops & Metrics

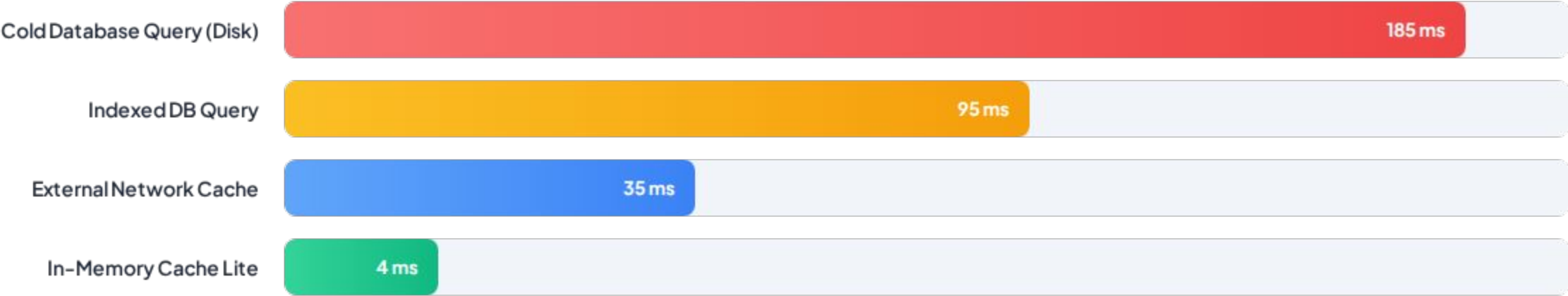
Engineered granular deletion mechanisms alongside atomic hit/miss telemetry tracking for rigorous performance evaluation.



Phase 5 & 6: Reports & QA

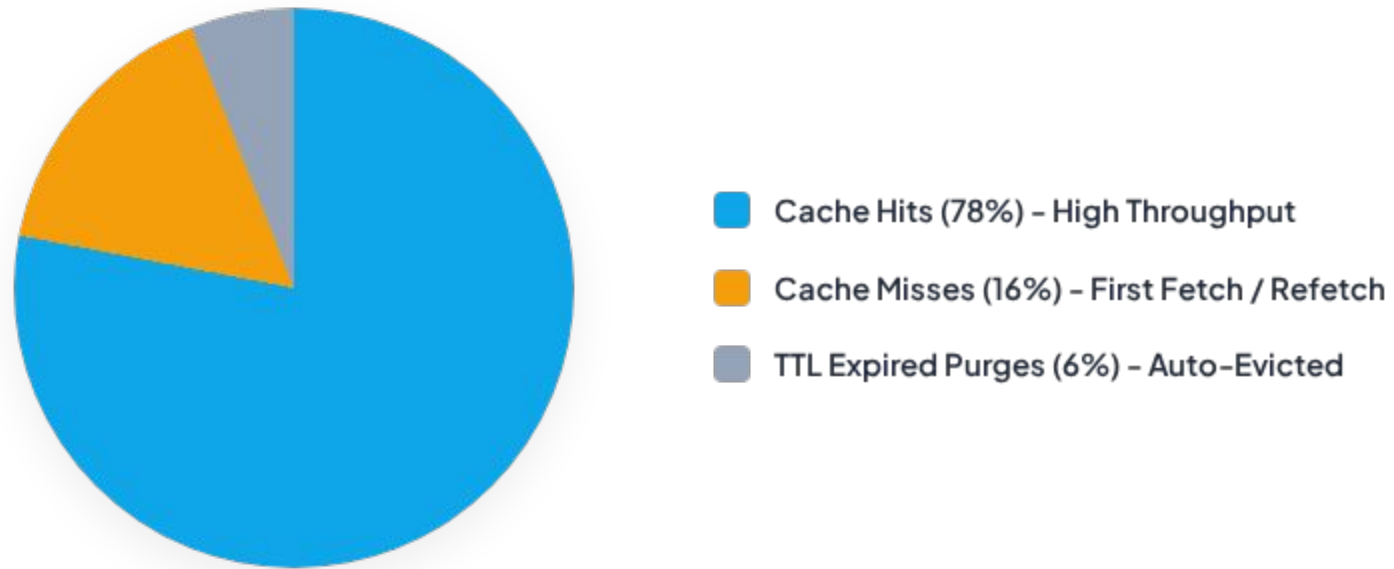
Constructed reporting utilities for operational insights and executed unit test scenarios simulating extreme concurrency.

Latency Benchmark Comparison



*In-Memory Cache with TTL Lite achieves sub-5ms retrieval latency, providing up to **97.8% latency reduction** over traditional database access paths.*

Cache Request Distribution



The 78% cache hit ratio confirms superior retrieval efficiency while automatic TTL pruning ensures zero stale state retention.

Outcomes & Key Learnings

- ✓ **Demonstrated Memory Efficiency:** Mastered dynamic heap memory management, pointer safety, and automated garbage collection routines in JavaScript.
- ✓ **Full-Stack Mastery:** Gained end-to-end expertise spanning React frontends, Express APIs, MongoDB databases, and micro-caching layers.
- ✓ **Industry Best Practices:** Applied clean architecture principles, modular separation of concerns, Git workflow management, and defensive programming.
- ✓ **Academic & Professional Growth:** Successfully fulfilled the B.Tech internship requirement with a high-performance capstone evaluated by EduSkills Academy.

Thank You

Questions & Feedback are

Welcome



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